

INVESTIGATION REPORT

Issue 1005: Outer loop ending inside the inner loop

openhwgroup/cv32e40p

|                |                                                         |
|----------------|---------------------------------------------------------|
| Document       | RTL-TRIAGE-CV32E40P-1005                                |
| Disposition    | Waiver, with a stated software rule                     |
| Record date    | 2026-09-18                                              |
| Classification | Internal working document. Not a verification sign-off. |

1 Status

|               |                                                   |
|---------------|---------------------------------------------------|
| Issue         | openhwgroup/cv32e40p#1005                         |
| Issue type    | Coverage hole referred for disposition            |
| Affects       | cv32e40p                                          |
| Configuration | COREV_PULP=1, FPU=0 (RTL master 6033d2b)          |
| Disposition   | Waiver, with a stated software rule               |
| Severity      | Not assessed                                      |
| Agreement     | Same disposition as the independent human record. |
| Evidence      | 9 cited, 7 present, 2 not retained                |

Disposition meaning. The code is unreachable only while software obeys an explicitly stated rule. The rule is part of the disposition and is recorded in the Action section.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

cv32e40p\_controller.sv:640 is 'if (hwlp\_end1\_eq\_pc && hwlp\_counter1\_gt\_1)' inside 'if (is\_hwlp\_body)' in the DECODE default arm. The uncovered condition is hwlp\_end1\_eq\_pc==1 with hwlp\_counter1\_gt\_1==0 while is\_hwlp\_body==1. Since counter1 <= 1 kills HWLoop1's own term, is\_hwlp\_body must come from HWLoop0 (start0 <= pc, end0 >= pc+4, counter0 > 1), so HWLoop1's end register must equal pc+4 with pc inside HWLoop0's body: the outer loop's end sits inside the inner loop, the same nesting violation as #1009 (mirror of line 630, which is covered because loop0 nested in loop1 is legal). Hardware does not check this: the cover engine reaches the exact condition in 7 cycles with an illegal register set (and address wrap at the top of memory). Under the PULP nesting rule (end1 >= end0+8, start0 >= start1) it is impossible: z3 unsat with the rule (also unsat for disjoint sequential loops), sat without it (model: end1 = pc+4 inside loop0, counter1=0). Formal proof under the invariant: a first attempt with only the #1009 register-content rule was defeated by a 32-bit wrap witness (start0 > end0 at 0xfffffff, kept as evidence); adding well-formed loops (start < end when active) and no address wrap, k-induction (k=12) proves the condition unreachable. abc pdr did not converge. Caveat as in #1009: a stale HWLoop1 end register (counter1 = 0 after a finished loop) lying inside a later HWLoop0 body also reaches the arm with sequential, non-nested loops, so the waiver invariant must be read on register contents. Disposition matches the 2024 waiver but replaces 'could maybe be proven with SLEC' by an open-source proof under a stated invariant.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

## 4 Implications

*Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.*

## 5 Action

### Waiver text, to be applied to the coverage exclusion

cv32e40p\_controller.sv:640 condition hwlp\_end1\_eq\_pc && !hwlp\_counter1\_gt\_1 (with is\_hwlp\_body): reachable only if HWLoop1's end register lies inside an active HWLoop0 body (end1 in [start0+4, end0]), which the PULP HWLoop nesting rule (HWLoop0 inner, outer end >= inner end + 8) forbids. Under the invariant 'whenever hwlp\_counter[0] > 1, hwlp\_end\_addr[1] is not in [hwlp\_start\_addr[0], hwlp\_end\_addr[0]+4]' plus well-formed loops (start < end) and no 32-bit wrap, the condition is formally unreachable (k-induction k=12, runs/v1-cv32e40p-1005/formal/p\_line640\_inv2\_kind; z3 runs/v1-cv32e40p-1005/smt). Hardware does not enforce the rule (bounded cover reaches it in 7 cycles, runs/v1-cv32e40p-1005/formal/cover). Waive as software-illegal with that register-content invariant (it also covers stale HWLoop1 registers). Line 630 is the legal mirror and stays covered.

### Checker used

checker\_1005.sv

## 6 Evidence

| Ref | Type           | Path, relative to the run directory              | File     | Result recorded                                                                                                                                                               |
|-----|----------------|--------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1  | proof          | formal/cover/sby.log                             | present  | cover PASS: line 630 and 640 taken arms at step 7; the hole condition (end1==pc+4, counter1<=1, is_hwlp_body) at step 7 -> reachable by hardware with illegal/stale registers |
| E2  | counterexample | formal/cover/job/engine_0/trace0.vcd             | not kept | hole witness: pc=0xffffffff inside loop0 [0xffffffff, 0x0 (wrapped)] cnt0=0xffff, end1=0x0=pc+4, cnt1=0 (cover_hole_line640_hw_only.png)                                      |
| E3  | smt-model      | smt/with_invariant.out                           | present  | z3 unsat: line-640 condition + is_hwlp_body impossible under end1 >= end0+8, start0 >= start1, no wrap                                                                        |
| E4  | smt-model      | smt/with_disjoint.out                            | present  | z3 unsat also for disjoint sequential loops (loop1 after loop0, non-empty bodies)                                                                                             |
| E5  | smt-model      | smt/without_invariant.out                        | present  | z3 sat without the rule: end1 = pc+4 inside an active loop0, cnt1 = 0                                                                                                         |
| E6  | proof          | formal/p_line640_inv2_kind/sby.log               | present  | k-induction (smtbmc yices, k=12) PASS under: end1 not in [start0, end0+4] while counter0>1; start<end for active loops; no address wrap                                       |
| E7  | counterexample | formal/p_line640_inv_kind/job/engine_0/trace.vcd | not kept | v2 attempt (register rule only) FAIL: wrap witness start0=0xffffffff > end0=0xffffffff8; led to the well-formedness assumptions                                               |
| E8  | proof          | formal/p_line640/sby.log                         | present  | negative result kept: unconstrained abc pdr TIMEOUT (property false without invariant)                                                                                        |
| E9  | proof          | formal/p_line640_inv/sby.log                     | present  | negative result kept: abc pdr with invariant did not converge; k-induction used                                                                                               |

*Paths are relative to runs/v1-cv32e40p-1005. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.*

### What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- smt-model. A satisfying assignment from the SMT solver for the stated query.

Artefact(s) E2, E7 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

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Limitations and scope

- The waiver holds only while software obeys the rule stated in Action. If software can violate that rule, the waiver does not hold and the code is reachable.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

|                     |                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Run directory       | runs/v1-cv32e40p-1005                                                                                                                      |
| Title source        | Title and configuration read back from reports/issues/cv32e40p_issue_1005.pdf, the previously published report for this issue.             |
| Inputs shown        | title, body(links stripped), labels                                                                                                        |
| Human record        | Dolphin Design 2024 (issue body): waived in v2 by test reasoning (HWLoop1 cannot be nested in HWLoop0); 'could maybe be proven using SLEC' |
| Assessed confidence | 0.75. This is the investigator's own estimate. It is not a measurement and not a reliability figure.                                       |
| Iterations          | 3                                                                                                                                          |

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